MILITARY SPECIFICATION

NAVIGATION SET, RADIO AN/ARN-87 (V)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall function as an airborne VHF navigation system.

1.2 Classification - The Radio Navigation Set AN/ARN-87 (V) shall consist of the following items:

Items	Type Designation	Appl. Para.
Receiver, Radio	R-1185(*)/ARC-101	3.4.1
Converter, Navigation Set	CV-2059(*)/ARN-87 (V)	3.4.2
Control, Radio Set	C-6842(*)/ARN-87 (V) through C-6845(*)/ARN-87 (V)	3.4.3
Mounting (Receiver)	MT-3593(*)/ARN-87 (V)	3.4.4
Mounting (Converter)	MT-3594(*)/ARN-87 (V)	3.4.5
Mounting (Receiver and Converter)	MT-3595(*)/ARN-87 (V)	3.4.6
Mounting (Converter)	MT-3596(*)/ARN-87 (V)	3.4.7
Power Supply (Receiver DC)	PP-2448(*)/ARC-84	3.4.8
Power Supply (Receiver AC)	PP-2688(*)/ARC-84	3.4.9

FSC 5826

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for
MIL-R-81217	Radio Set AN/ARC-101

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

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PUBLICATIONS

Radio Technical Commission for Aeronautics (RTCA)

DO-114(Paper 48-62)	Minimum Performance Standards - Airborne VOR Receiving Equipment Operating within the Radio-Frequency Range of 108-118 Megacycles.
DO-116(Paper 21-63)	Minimum Performance Standards - Airborne ILS Glide Slope Receiving Equipment.
DO-131	Minimum Performance Standards - Airborne ILS Localizer Receiving Equipment.

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120. Copies of RTCA Documents may be obtained from Radio Technical Commission for Aeronautics, 2000 K Street, N. W., Washington, D. C., 20006.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

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(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein, shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.3)

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied as outlined in Specification MIL-E-4682.

3.2 Design and Construction - The equipment shall conform with all applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.2.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed the sum of the individual weights specified for the units used.

3.2.2 Operating Life

3.2.2.1 Operating Life - The equipment shall have a minimum total operating life of 40,000 hours with reasonable servicing and replacement of parts. Information on parts requiring replacement within this interval and the wearout life of such parts as determined by the contractor shall be submitted to the procuring agency.

3.2.2.2 Reliability in Mean Time Between Failures (MTBF) - The equipment shall have 100 hours of mean (operating) time between failures when tested and accepted under the requirements of 4.4.3.

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3.2.3 Cabling and Connections

3.2.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400 except for the requirement of maintaining a specific minimum of spare (unused) contacts per connector.

3.2.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088, except as stated herein. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements, and provided the assembly of the cable to its plugs may be easily accomplished. External cables shall not be supplied as part of the equipment, but that portion of connectors to which the cables attach shall be furnished as a part of the mounting base for the transmitter, receiver and converter. The external cable shall include a Cannon Electric Company part no. KOA6-21-50SN connector or approved equivalent for proper mating to the control panel connector supplied as part of the equipment. The external cabling shall include primary power control circuitry with relay solenoid return to ground being controlled by a switch within the control unit.

3.2.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.5 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.2.6 Interference Control - The generation of radio interference by the equipment including receiver and controls and the vulnerability of the equipment including receiver and controls to radio interference shall meet the requirements of RTCA Documents DO-114, DO-116 and DO-131.

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3.2.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance, but these facilities need not necessarily be available on the outer case. In location and mounting of component parts, consideration shall be given to accessibility for ease in maintenance of the equipment. A removable type of plug-in construction shall be incorporated for any subassembly that might otherwise be difficult to service properly.

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25° C ±5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ±1.0V AC or 27.5 ±0.5V DC

3.2.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class I equipment, except as modified herein:

3.2.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of MIL-E-5400.

3.2.10.2 Altitude - The equipment shall operate satisfactorily at any altitude between sea level and 55,000 feet.

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3.2.10.3 Temperature - The equipment shall operate satisfactorily in any ambient temperature within the range of -40° C to +55°C. The equipment shall be capable of being stored for an extended period of time in any ambient temperature within the range of -65°C to +85°C.

3.2.11 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under extreme service conditions.

3.2.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from either of the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required for the receiver, converter, mountings and control shall not exceed the specified amounts.

(1) AC power (Single Phase) 115V, Category "B" 156 VA

or

(2) DC power 28V Category "B" 9.5 Amps (13 Amps during frequency channeling)

3.2.12.1 Input Power Connection - Input power connectors shall be as specified in paragraph 3.4.4.6 and the relay control of primary power shall be incorporated in external wiring as specified in paragraph 3.2.3.2.

3.2.13 Duty Cycle - The receiver and converter shall be capable of continuous operation.

3.3 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified. Performance shall be in accordance with approved test procedures, unless otherwise indicated.

3.3.1 Operation - The equipment shall operate as an airborne VHF navigation system or, when integrated with Radio Set AN/ARC-101, as a VHF communication navigation system. The receiver shall provide reception of VHF signals on 880 crystal-controlled communications, visual omnirange (VOR) and localizer (LOC) channels, spaced 50 KC apart over the 108.00 to 151.95-megacycles range. Channel selection shall be accomplished by electric control signals from a remote control unit associated with the equipment. The converter shall accept audio signals from the associated receiver when the receiver is in any VOR or LOC channel in the 108.00 to

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117.9 megacycles band; and, in conjunction with appropriate indicator instruments, the converter shall convert the VOR/LOC audio into the signals necessary to operate the VOR/LOC visual indicator instruments to provide VOR/LOC course information to the pilot or navigator. For detailed operation of the transmitter and receiver see MIL-R-81217.

3.3.2 Channels - The receiver shall receive 880 channels, spaced 50 KC apart, over a frequency of 108.00 to 151.95 MC band.

3.3.3 Internal Power Supplies - The receiver shall have provisions for operating with either ac or dc plug-in power supplies.

3.3.4 Remote Control - Receiver squelch, volume and operating frequency shall be remotely controlled from the control unit.

3.4 Detail Requirements

3.4.1 Receiver, Radio R-1185 (*)/ARC-101 - The receiver shall meet the requirements of MIL-R-81217 (WP).

3.4.2 Converter, Navigation Set CV-2059/ARN-87(V) - The converter shall meet the following requirements.

3.4.2.1 Function - The converter shall accept audio signals from the associated receiver when the receiver is in any VOR or LOC channel in the 108.0 to 117.9 megacycle band and shall, with appropriate indicator instruments, convert the VOR/LOC audio into signals necessary to provide the VOR/LOC course information to the pilot or navigator.

3.4.2.2 Form Factor - The converter form factor and dimensions shall be in accordance with Figure 1.

3.4.2.3 Weight - The weight of the converter shall not exceed 9.0 pounds.

3.4.2.4 Contents of Unit - The converter shall contain the necessary circuitry to perform the functions of 3.4.2.1.

3.4.2.5 Controls - The following adjustable controls shall be accessible with the converter removed from its case:

3.4.2.5.1 LOC. BAL. (Localizer Balance) - The localizer balance potentiometer shall adjust the standard centered localizer input.

3.4.2.5.2 LOC. SENS. (Localizer Sensitivity) - The localizer sensitivity potentiometer shall adjust the current to the required value.

3.4.2.5.3 LOC. FLAG (Localizer Flag) - The localizer flag potentiometer shall adjust the standard localizer signal.

3.4.2.5.4 REF. FLAG, OMNI FLAG SENS and VAR. FLAG - The reference flag, omni flag sensitivity and variable flag potentiometers shall adjust the VOR reference phase and variable phase and flag amplifier circuits.

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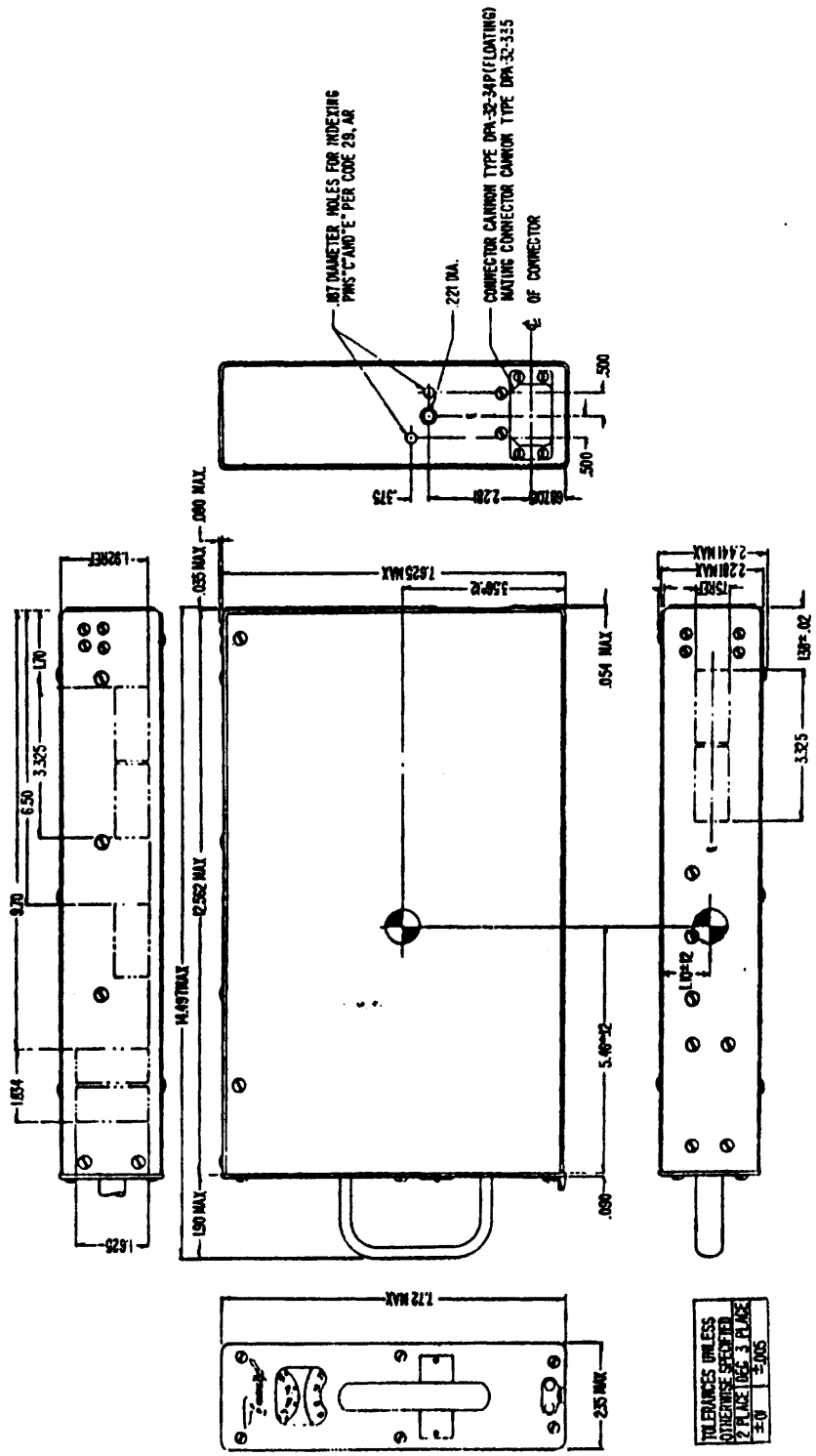


Figure 1. Converter, Navigation Set, CV-2059/ARN-87 (V), Outline Dimensions.

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3.4.2.5.5 MAN. SENS. (Manual Sensitivity) - The manual sensitivity potentiometer sets the correct course width.

3.4.2.5.6 MAN. BAL. (Manual Balance) - The manual balance potentiometer shall balance the reciprocal error.

3.4.2.5.7 AUTO ZERO and AUTO TRACKING - The automatic zero and automatic tracking potentiometer shall electrically align the OBI at 10-degree and 65-degree positions.

3.4.2.5.8 SERVO CHECKER SENS - The servo checker sensitivity potentiometer shall adjust the slot width in the servo checker circuit.

3.4.2.5.9 VOR CHECK ZERO - The VOR check zero potentiometer shall adjust the variable phase signal for correct phasing when unit is in the test mode.

3.4.2.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
P1	Cannon Electric Co. part no. DPA-32-34P (Floating) or approved equivalent	Control and power inlet and converter outputs.

3.4.2.7 Converter Performance Requirements

3.4.2.7.1 Visual Omnirange and Localizer Course - The converter, when operating in conjunction with the R-1185(*)/ARC-101 (or equivalent) receiver and appropriate indicator instruments, shall provide VOR/LOC course information to the pilot or navigator.

3.4.2.7.2 Navigation Audio Signals - The converter shall accept navigation audio signals from the associated receiver in any omni or localizer channel in the 108.00 to 117.9 megacycles band.

3.4.2.7.3 Omni Channel - The navigation audio input in the omni channel shall be a 30-cps variable phase signal combined with a 9960-cps subcarrier (frequency modulated by a 30-cps reference phase signal).

3.4.2.7.4 Localizer Channel - The navigation audio input in the localizer channel shall be a 90/150-cps composite signal.

3.4.2.7.5 NAV Audio - The converter shall convert the NAV audio into the signals necessary to operate the VOR/LOC visual indicator instruments.

3.4.2.7.6 Localizer Course - The converter signals shall enable visual presentation of localizer course as presented on a Bendix Type MN-97 Omni-Mag Indicator, INA-21 Course Indicator, or equivalent flight path direction indicator (FPDI).

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3.4.2.7.7 Manual Course - The converter signals shall enable visual presentation of VOR (manual) course as presented on a Bendix Type MN-97 Omni-Mag Indicator, INA-21 Course Indicator, or equivalent FPDI.

3.4.2.7.8 VOR Bearing - The converter signals shall enable visual presentation of heading-sensitive (automatic) VOR bearing as presented on a Bendix Type MN-72 dual-pointer Radio Magnetic Indicator (RMI) or equivalent.

3.4.2.7.9 Error Indication - The converter shall provide a pushbutton-controlled checking circuit (VOR-CHECK) to determine and indicate errors that occur in the automatic or manual VOR circuits. This circuit shall indicate that an error exists and indicate the amount of error.

3.4.2.7.10 Transient Voltage Protection - The converter shall not be damaged by transient spikes of the following amplitude and duration:

Amplitude	Duration
600 volts	1 microsecond
300 volts	10 microseconds
150 volts	300 microseconds
80 volts	100 milliseconds

3.4.2.7.11 Short Circuit Protection - The converter shall not be damaged by short-circuits of the regulated +21.5 volts dc of indefinite period of time.

3.4.3 Control, Radio Set C-6842/ARN-87(V) through C-6845/ARN-87(V) - The controls shall meet the following requirements:

3.4.3.1 Function - The control panels shall provide the following remote control functions: either frequency selection of the transmitter and receiver (C-6842, 6844) or receiver only (C-6843, 6845), and squelch and volume on-off control of the receiver. Also, the control panels shall have a VOR-CHECK switch which operates a test circuit to quick-check VOR operation.

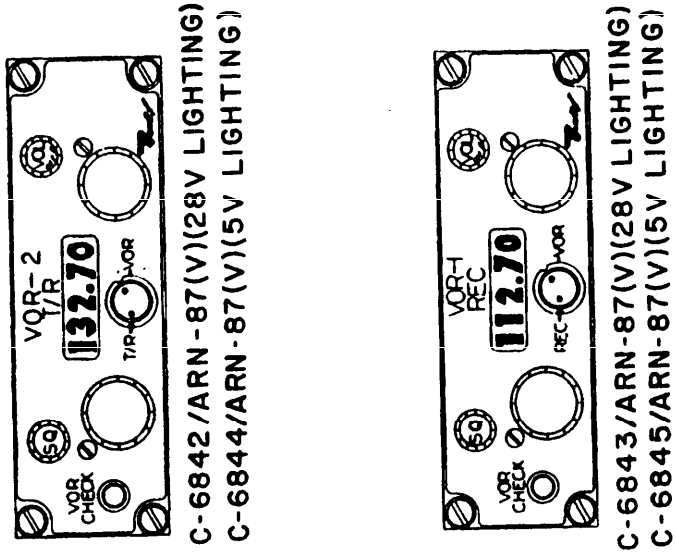
3.4.3.2 Form Factor - The control panel form factor and dimensions shall be in accordance with Figure 2 and the applicable requirements of Specification MIL-C-6781 for Type I control panels.

3.4.3.3 Weight - The weight of the control panel shall not exceed 1.6 pounds.

3.4.3.4 Contents of Control Panel

3.4.3.4.1 Whole-Megacycle Frequency Selection Circuit - The control panel shall have a front panel whole-megacycle frequency control knob controlling a direct-readout-counter-type frequency indicator. The frequency indicator shall be

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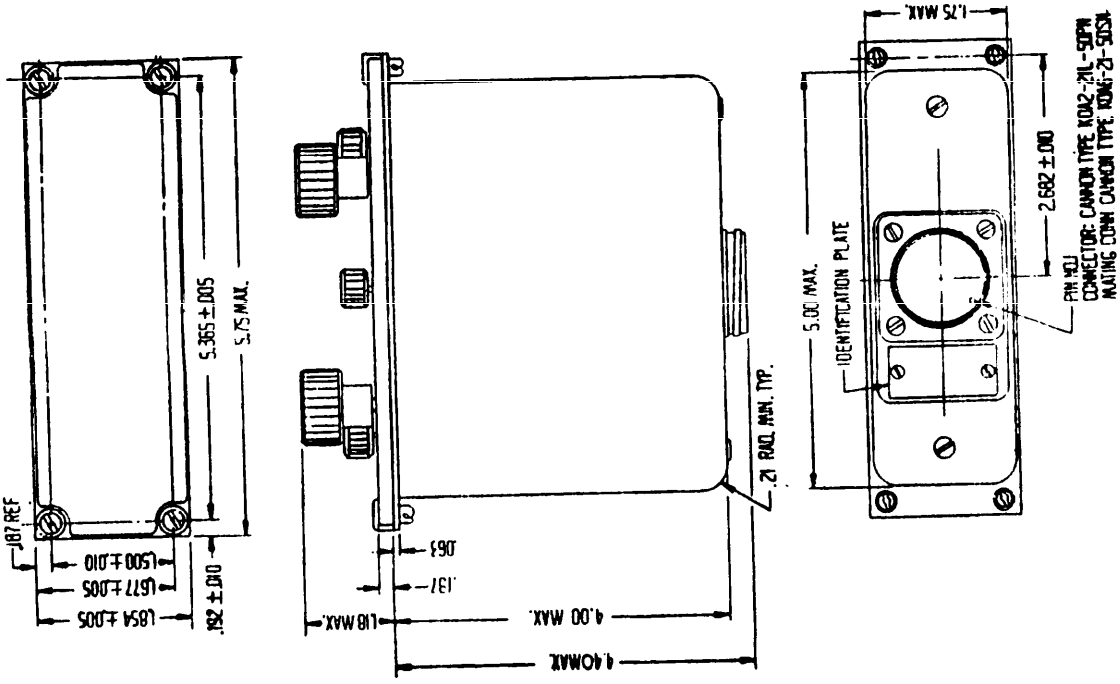


Figure 2. Control, Radio Set, C-6842/ARN-87 (V), C6843/ARN-87 (V) C-6844/ARN-87 (V) and C-6845/ARN-87 (V), Outline Dimensions

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numbered from 108 to 151 and the numbers shall appear in increasing sequence as the knob is rotated clockwise. The switch shall select the whole-megacycle operating frequency of the transmitter and receiver to agree with the frequency appearing on the frequency indicator.

3.4.3.4.2 Fractional-Megacycle Frequency Selection Circuit - The control panel shall have a front panel fractional-megacycle frequency control knob controlling a direct-readout-counter type frequency indicator. The frequency indicator shall be numbered from .00 to .95 in increments of .05, and the numbers shall appear in increasing sequence as the knob is rotated clockwise. The switch shall select the fractional-megacycle operating frequency of the transmitter and the receiver to agree with the frequency appearing on the frequency indicator.

3.4.3.4.3 Volume and Power Control Circuit - The control panel shall have a front panel VOL control incorporating an on-OFF switch which shall be OFF in the extreme counterclockwise position and otherwise on.

3.4.3.4.4 Squelch Control Circuit - The control panel shall have a front panel SQ. (squelch) control potentiometer for adjustment of the squelch level in the receiver.

3.4.3.5 Location of Controls -

3.4.3.5.1 The whole-megacycle frequency control knob shall be mounted on the left side of the front panel.

3.4.3.5.2 The fractional-megacycle frequency control knob shall be mounted on the right side of the front panel.

3.4.3.5.3 A VOL control on-OFF knob shall be mounted above the fractional-megacycle frequency control knob.

3.4.3.5.4 A squelch control knob shall be mounted above the whole-megacycle frequency control knob.

3.4.3.6 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J1	Cannon Electric Co. part no. KOA2-21L-50PN or approved equivalent	Power inlet and control output

3.4.4 Mounting MT-3593/ARN-87(V) - The mounting shall meet the following requirements:

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3.4.4.1 Function - The mounting shall function as a support base for R-1185(*)/ARC-101 receiver.

3.4.4.2 Form Factor - The mounting form factor and dimensions shall be in accordance with Figure 3.

3.4.4.3 Weight - The weight of the mounting shall not exceed 1.6 pounds.

3.4.4.4 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J101	Cannon Electric Co. part no. DPA-29C1-33S or approved equivalent	Receiver output and control and power inlet

3.4.4.5 Performance Requirements - The mounting shall be arranged to furnish all support required by the receiver and shall contain such vibration isolators as required.

3.4.5 Mounting MT-3594/ARN-87(V) - The mounting shall meet the following requirements:

3.4.5.1 Function - The mounting shall function as a support base for CV-2059/ARN-87(V) converter on other than carrier operated aircraft.

3.4.5.2 Form Factor - The mounting form factor and dimensions shall be in accordance with Figure 3.

3.4.5.3 Weight - The weight of the mounting shall not exceed 1.6 pounds.

3.4.5.4 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J101	Cannon Electric Co. part no. DPA-32-33S or approved equivalent	Control and power inlet and converter outputs

3.4.5.5 Performance Requirements - The mounting shall be arranged to furnish all support required by the converter and shall contain such vibration isolators as required.

3.4.6 Mounting MT-3595/ARN-87(V) - The mounting shall meet the following requirements:

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TABLE I

MOUNTING		EQUIPMENT USED WITH		CARRIER TYPE CONNECTOR		INDEX PINS		DIMENSIONS		
TYPE NO.	WEIGHT	TYPE NO.	WEIGHT	POSITION	TYPE NO.	CODE NO.	LOCATION	A	B	C
MT-3593/ARN-87 (V)	1.6 LBS	R-1185/ARC-101	9.0 LBS	UPRIGHT	DPA-2301-335	28	B & F	6.15	3.86	1.32
MT-3594/ARN-87 (V)	1.6 LBS	CV-2059/ARN-87(V)	9.2 LBS	UPRIGHT	DPA-32-335	29	C & E	7.12	3.44	1.32

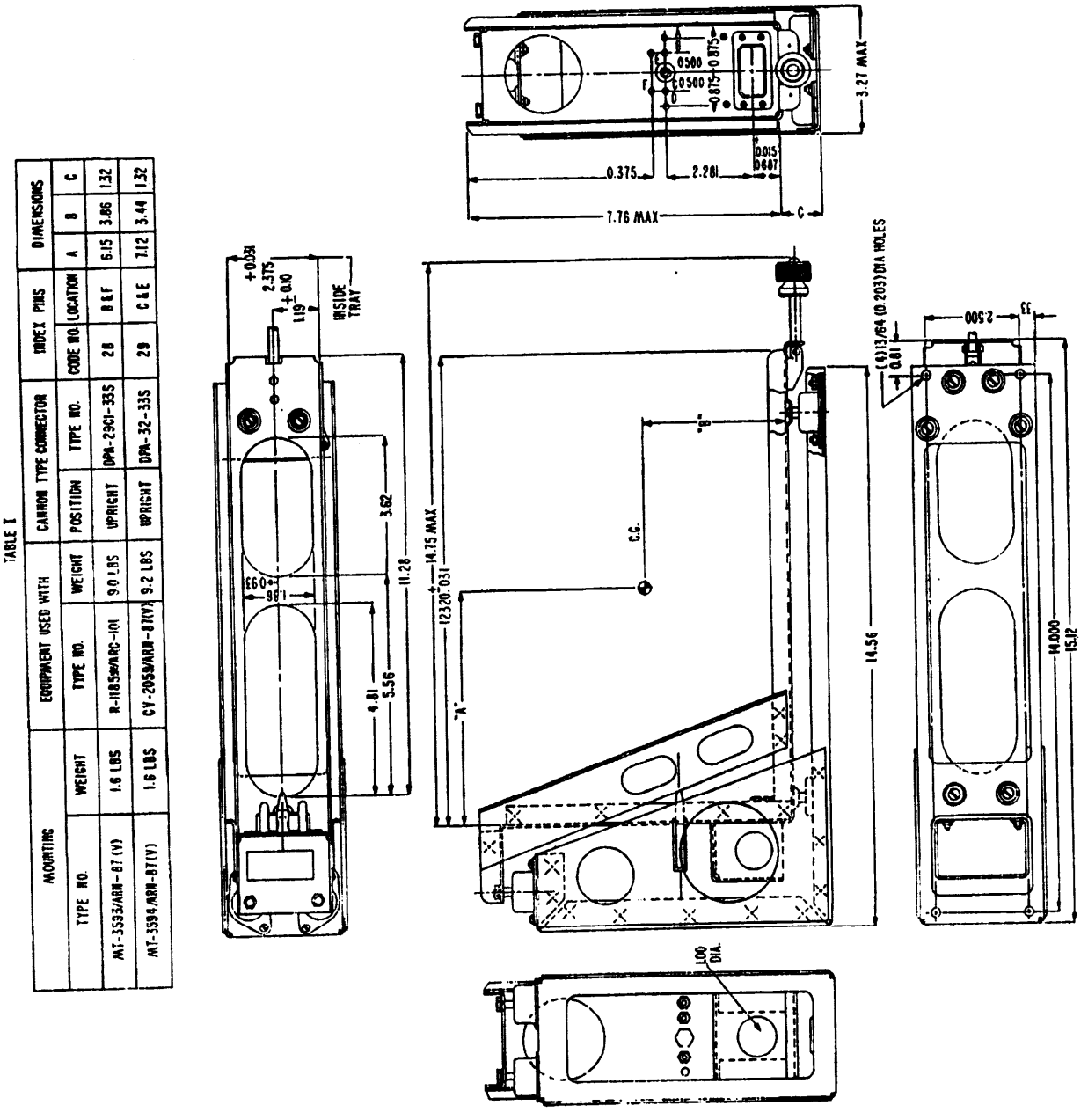


Figure 3. Mounting, MT-3593/ARN-87 (V) and MT-3594/ARN-87 (V) Outline Dimensions

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3.4.6.1 Function - The mounting shall function as a support base for R-1185(*)/ARC-101 receiver and CV-2059/ARN-87(V) converter.

3.4.6.2 Form Factor - The mounting form factor and dimensions shall be in accordance with Figure 4.

3.4.6.3 Weight - The weight of the mounting shall not exceed 2.1 pounds.

3.4.6.4 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J101	Cannon Electric Co. part no. DPA-29C1-33S or approved equivalent	Control and power inlet and receiver outputs
J102	Cannon Electric Co. part no. DPA-32-33S or approved equivalent	Control and power inlet and converter outputs

3.4.6.5 Performance Requirements - The mounting shall be arranged to furnish all support required by the receiver and converter and shall contain such vibration isolators as required.

3.4.7 Mounting MT-3596/ARN-87(V) - The mounting shall meet the following requirements:

3.4.7.1 Function - The mounting shall function as a support base for CV-2059/ARN-87(V) converter on carrier operated aircraft.

3.4.7.2 Form Factor - The mounting form factor and dimensions shall be in accordance with Figure 5.

3.4.7.3 Weight - The weight of the mounting shall not exceed 1.6 pounds.

3.4.7.4 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
J101	Cannon Electric Co. part no. DPA-32-33S or approved equivalent	Control and power inlet and converter outputs

3.4.7.5 Performance Requirements - The mounting shall be arranged to furnish all support required by the converter and shall contain such vibration isolators as required.

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TABLE I

MOUNTING		EQUIPMENT USED WITH			CANNON TYPE CONNECTOR			INDEX PINS		DIMENSIONS	
TYPE	WEIGHT	POSITION	TYPE NO.	WEIGHT	LOCATION	POSITION	TYPE NO.	CODE NO.	LOCATION	A	B
MT-3595/ARN-87 (V)	2.1 LBS	A	R-1185*/ARC-101	9.0 LBS.	POSITION A	UPRIGHT	DPA-29C1-33S	28	B1, F1	5.06	1.36
		B	CV-2059/ARN-87(V)	9.2 LBS.	POSITION B	UPRIGHT	DPA-32-33S	29	C2, E2		

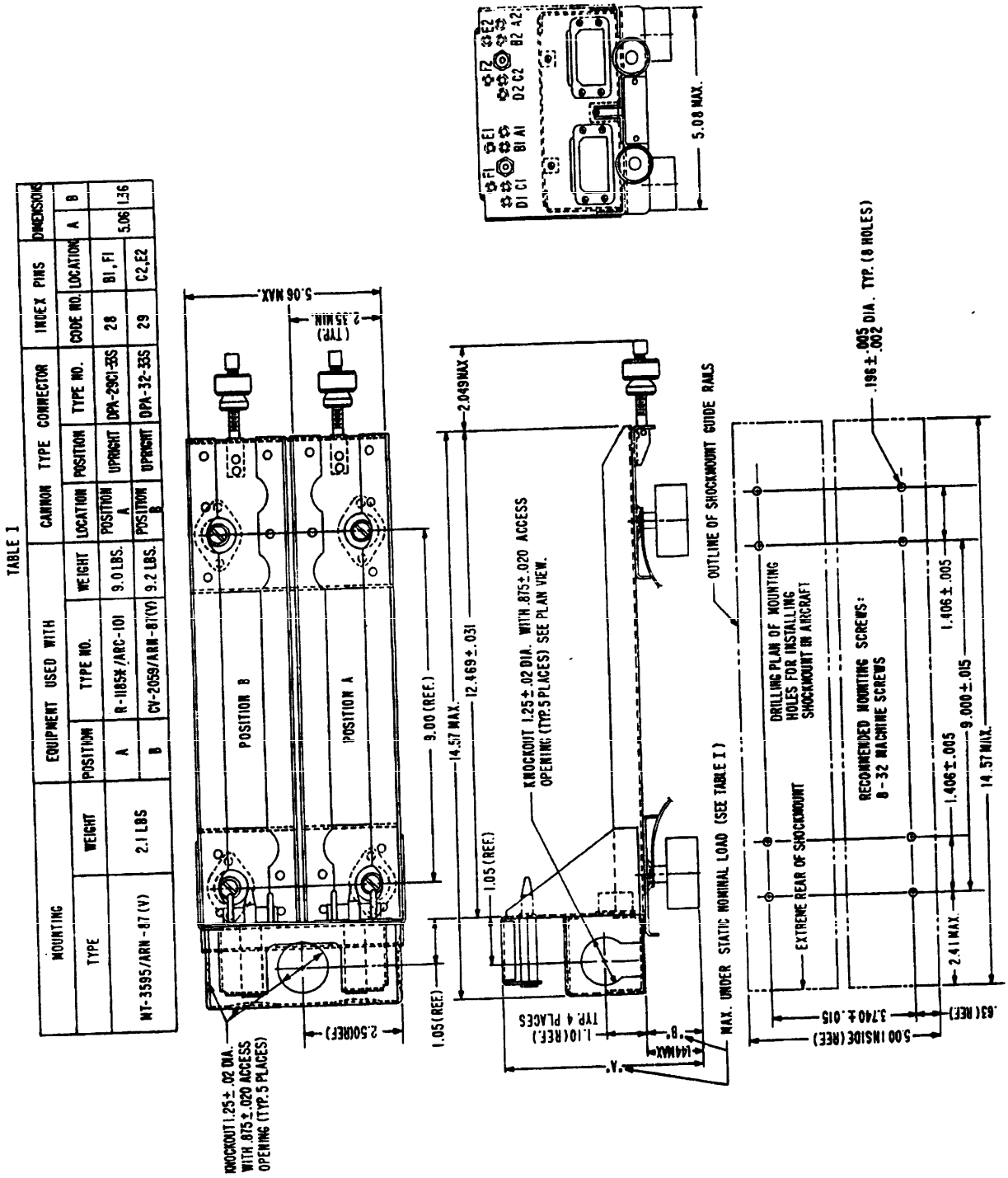


Figure 4. Mounting, MT-3595/ARN-87 (V) Outline Dimensions

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TABLE I

MOUNTING		EQUIPMENT USED WITH			CANNON TYPE CONNECTOR		INDEX PINS	
TYPE NO.	WEIGHT	TYPE NO.	WEIGHT	POSITION	TYPE NO.	CODE NO.	LOCATION
MT-3596/ARN-87(V)	1.6 LBS	CV-2059/ARN-87(V)	9.2 LBS	UPRIGHT	DPA-32-33S	29	C&E

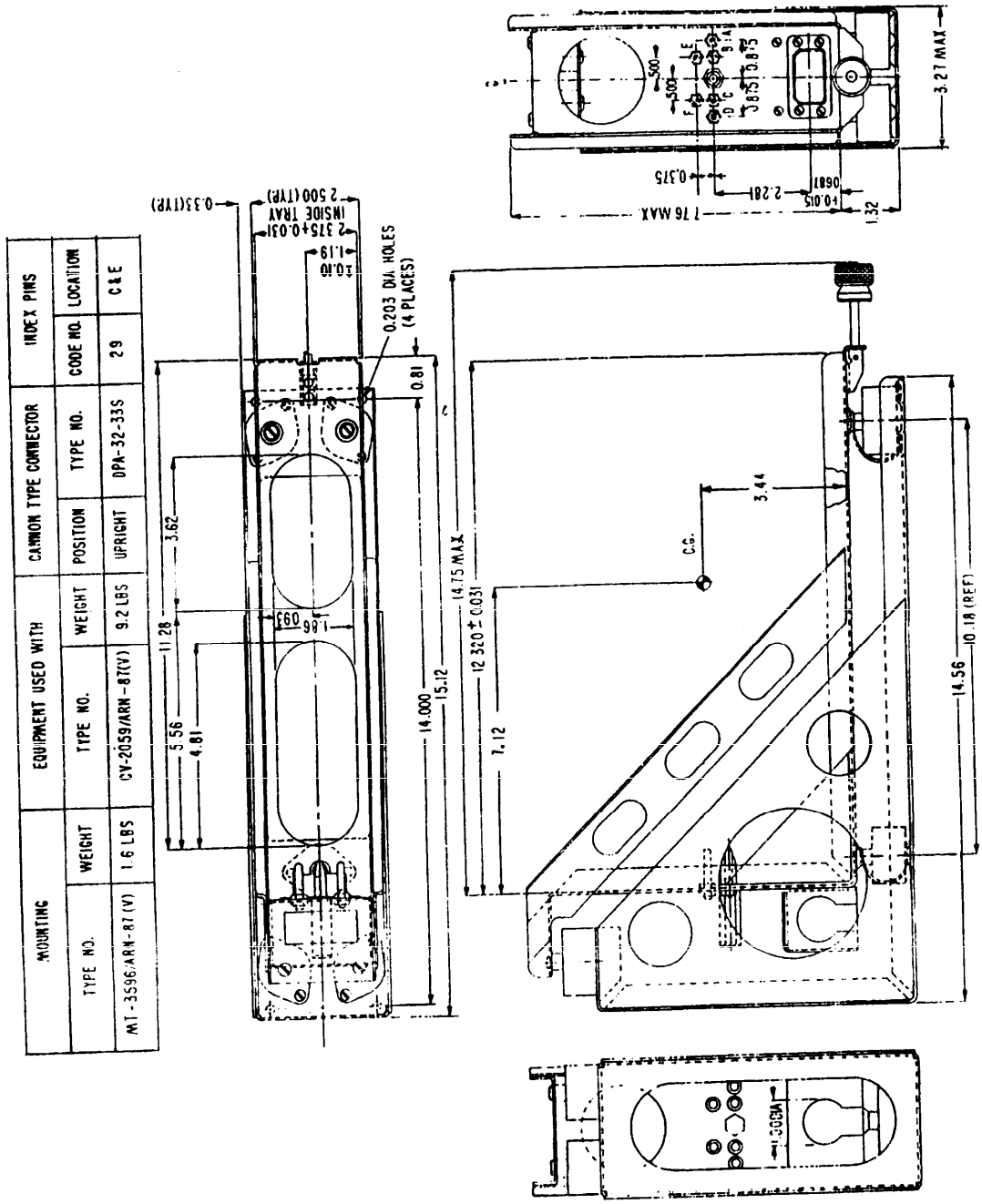


Figure 5. Mounting, MT-3596/ARN-87 (V) Outline Dimensions

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3.4.8 Power Supply PP-2448/ARC-84 - The power supply for Receiver DC operation shall meet the requirements of MIL-R-81217 (WP).

3.4.9 Power Supply PP-2688/ARC-84 - The power supply for Receiver AC operation shall meet the requirements of MIL-R-81217 (WP).

3.5 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (See paragraph 6.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall

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include environmental tests in accordance with the procedures of Specification MIL-T-5422 and interference tests necessary to determine compliance with the requirements herein.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

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4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run In Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's

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The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling Tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action

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necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by Specification MIL-R-23094. Equipments selected for reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Reliability Procedure - Reliability Procedure II from Specification MIL-R-23094 shall be used.

4.4.3.2 Test Level - The Test Level shall be Test Level III from Table I of Specification MIL-R-23094. During each cycle, the equipment shall be operated during the entire Heating Period shown in Table I of Specification MIL-R-23094.

4.4.3.3 Accept-Reject Criteria - Figure 15 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria.

4.4.3.4 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated four hours during each cycle.

4.4.3.5 Performance Characteristics to be Measured - The performance characteristics to be measured shall be as specified in 6.11 and shall be measured at least once during each operating portion of the cycle, at a random time during the cycle. The equipment shall operate on its normal duty cycle during the test.

4.4.3.6 Failure Criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the requirements of paragraph 4.4.3.5 at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.3.7 Preventive Maintenance - During the period of the tests, no preventive maintenance or readjustment of any controls which are inaccessible during flight may be performed.

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4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the pre-production tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity

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of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

4.5.4 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

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4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-R-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available such procedures shall be used. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V, 400 ± 20 cps AC

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as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance, or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Non-repairable Subassemblies - As a general rule, non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 The parentheses (*), when used in the type designation, may be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

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(a) Title, number, and data of this specification.

(b) Selection of applicable levels of packaging and packing (see 5. 1).

(c) Design Data

(d) Type of control panel, mounting and receiver power supply required

6. 7 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6. 8 Standard Test Signals - The RF input signal at 1000 μv shall be as follows:

6. 8. 1 Standard Audio Test Signal - An RF carrier amplitude modulated 30% at 1000 cps.

6. 8. 2 Standard VOR Test Signal - An RF carrier amplitude modulated simultaneously (1) 30% by a 9960 cps subcarrier which is, in turn, frequency modulated at a deviation ratio of 16 by a 30 cps "reference phase signal" and (2) 30% by a 30 cps "variable phase signal" which can be varied in phase with respect to the reference phase signal.

6. 8. 3 Standard Localizer Test Signal - An RF carrier amplitude modulated simultaneously with 90 ±0. 3% and 150 ±0. 3% cps signals so that the sum of their separate modulation percentages equals 40 ±2%.

6. 8. 4 Standard Localizer Centering Signal - A standard localizer test signal in which the difference in depth of modulation of the 90 and 150 cps signals is less than . 002 (. 1 db). NOTE: Difference in depth of modulation (ddm) is the percentage modulation depth of the larger signal minus the percentage modulation depth of the smaller signal divided by 100.

6. 8. 5 Standard Localizer Deviation Signal - A standard localizer test signal in which the difference in depth of modulation of the 90 and 150 cps signals is 0. 093 ±. 002 (4 ±. 1 db).

6. 9 Test Channels - The equipment shall be tested on the following channels:

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A. ILS localizer receiver - 110.1 mc

B. VOR receiver - 114.9 mc

6.10 Standard Deflection - Standard deflection shall be .6 of the center to full scale deflection of the deviation indicator. The equipment shall be adjusted to produce standard deflection when the input signal is a standard localizer deviation signal.

6.11 Performance Characteristics to be Measured -

6.11.1 Centering Characteristic - With a standard localizer centering signal, the deflection of the deviation indicator shall not change from that obtained under standard test conditions by more than 15% of standard deflection.

6.11.2 Deflection Characteristic - With standard localizer deviation signal, the deflection of the deviation indicator shall not change from that obtained under standard test conditions by more than 30% of standard deflection.

6.11.3 Deflection Linearity (VOR) - Shall meet the limits of standard test conditions. (Test with the minimum specified RF input level.)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-N-81405(WP) NAVIGATION SET, RADIO AN/ARN-87(V)

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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